

## 2 Functions of finite order

Let  $f$  be an entire function. If there exist a positive number  $\rho$  and constants  $A, B > 0$  such that

$$|f(z)| \leq Ae^{B|z|^\rho} \quad \text{for all } z \in \mathbb{C},$$

then we say that  $f$  has **an order of growth**  $\leq \rho$ . We define *the order of growth of  $f$*  as

$$\rho_f = \inf \rho,$$

where the infimum is over all  $\rho > 0$  such that  $f$  has an order of growth  $\leq \rho$ .

For example, the order of growth of the function  $e^{z^2}$  is 2.

**Theorem 2.1** *If  $f$  is an entire function that has an order of growth  $\leq \rho$ , then:*

- (i)  $n(r) \leq Cr^\rho$  for some  $C > 0$  and all sufficiently large  $r$ .
- (ii) If  $z_1, z_2, \dots$  denote the zeros of  $f$ , with  $z_k \neq 0$ , then for all  $s > \rho$  we have

$$\sum_{k=1}^{\infty} \frac{1}{|z_k|^s} < \infty.$$

*Proof.* It suffices to prove the estimate for  $n(r)$  when  $f(0) \neq 0$ . Indeed, consider the function  $F(z) = f(z)/z^\ell$  where  $\ell$  is the order of the zero of  $f$  at the origin. Then  $n_f(r)$  and  $n_F(r)$  differ only by a constant, and  $F$  also has an order of growth  $\leq \rho$ .

If  $f(0) \neq 0$  we may use formula (2), namely

$$\int_0^R n(x) \frac{dx}{x} = \frac{1}{2\pi} \int_0^{2\pi} \log |f(Re^{i\theta})| d\theta - \log |f(0)|.$$

Choosing  $R = 2r$ , this formula implies

$$\int_r^{2r} n(x) \frac{dx}{x} \leq \frac{1}{2\pi} \int_0^{2\pi} \log |f(Re^{i\theta})| d\theta - \log |f(0)|.$$

On the one hand, since  $n(r)$  is increasing, we have

$$\int_r^{2r} n(x) \frac{dx}{x} \geq n(r) \int_r^{2r} \frac{dx}{x} = n(r) [\log 2r - \log r] = n(r) \log 2,$$

and on the other hand, the growth condition on  $f$  gives

$$\int_0^{2\pi} \log |f(Re^{i\theta})| d\theta \leq \int_0^{2\pi} \log |Ae^{BR^\rho}| d\theta \leq C'r^\rho$$

for all large  $r$ . Consequently,  $\mathbf{n}(r) \leq Cr^\rho$  for an appropriate  $C > 0$  and all sufficiently large  $r$ .

The following estimates prove the second part of the theorem:

$$\begin{aligned} \sum_{|z_k| \geq 1} |z_k|^{-s} &= \sum_{j=0}^{\infty} \left( \sum_{2^j \leq |z_k| < 2^{j+1}} |z_k|^{-s} \right) \\ &\leq \sum_{j=0}^{\infty} 2^{-js} \mathbf{n}(2^{j+1}) \\ &\leq c \sum_{j=0}^{\infty} 2^{-js} 2^{(j+1)\rho} \\ &\leq c' \sum_{j=0}^{\infty} (2^{\rho-s})^j \\ &< \infty. \end{aligned}$$

The last series converges because  $s > \rho$ .

Part (ii) of the theorem is a noteworthy fact, which we shall use in a later part of this chapter.

We give two simple examples of the theorem; each of these shows that the condition  $s > \rho$  cannot be improved.

EXAMPLE 1. Consider  $f(z) = \sin \pi z$ . Recall Euler's identity, namely

$$f(z) = \frac{e^{i\pi z} - e^{-i\pi z}}{2i},$$

which implies that  $|f(z)| \leq e^{\pi|z|}$ , and  $f$  has an order of growth  $\leq 1$ . By taking  $z = ix$ , where  $x \in \mathbb{R}$ , it is clear that the order of growth of  $f$  is actually equal to 1. However,  $f$  vanishes to order 1 at  $z = n$  for each  $n \in \mathbb{Z}$ , and  $\sum_{n \neq 0} 1/|n|^s < \infty$  precisely when  $s > 1$ .

EXAMPLE 2. Consider  $f(z) = \cos z^{1/2}$ , which we *define* by

$$\cos z^{1/2} = \sum_{n=0}^{\infty} (-1)^n \frac{z^n}{(2n)!}.$$

Then  $f$  is entire, and it is easy to see that

$$|f(z)| \leq e^{|z|^{1/2}},$$

and the order of growth of  $f$  is  $1/2$ . Moreover,  $f(z)$  vanishes when  $z_n = ((n + 1/2)\pi)^2$ , while  $\sum_n 1/|z_n|^s < \infty$  exactly when  $s > 1/2$ .

A natural question is whether or not, given any sequence of complex numbers  $z_1, z_2, \dots$ , there exists an entire function  $f$  with zeros precisely at the points of this sequence. A necessary condition is that  $z_1, z_2, \dots$  do not accumulate, in other words we must have

$$\lim_{k \rightarrow \infty} |z_k| = \infty,$$

otherwise  $f$  would vanish identically by Theorem 4.8 in Chapter 2. Weierstrass proved that this condition is also sufficient by explicitly constructing a function with these prescribed zeros. A first guess is of course the product

$$(z - z_1)(z - z_2) \cdots,$$

which provides a solution in the special case when the sequence of zeros is finite. In general, Weierstrass showed how to insert factors in this product so that the convergence is guaranteed, yet no new zeros are introduced.

Before coming to the general construction, we review infinite products and study a basic example.

### 3 Infinite products

#### 3.1 Generalities

Given a sequence  $\{a_n\}_{n=1}^{\infty}$  of complex numbers, we say that the product

$$\prod_{n=1}^{\infty} (1 + a_n)$$

**converges** if the limit

$$\lim_{N \rightarrow \infty} \prod_{n=1}^N (1 + a_n)$$

of the partial products exists.

A useful necessary condition that guarantees the existence of a product is contained in the following proposition.